Faezeh Pasandideh E-Mail: faezeh.pasandideh@hshl.de

Stefan Henkler E-Mail: stefan.henkler@hshl.de

HOCHSCHULE HAMM-LIPPSTADT

1. Requirements Engineering
2. Requirement Specification
3. Approach: SysML
4. Discussion & Summary
5. Bibliography

- Fully automated baggage system
- 2 years delay (1993 → 1995)
- $500M extra cost



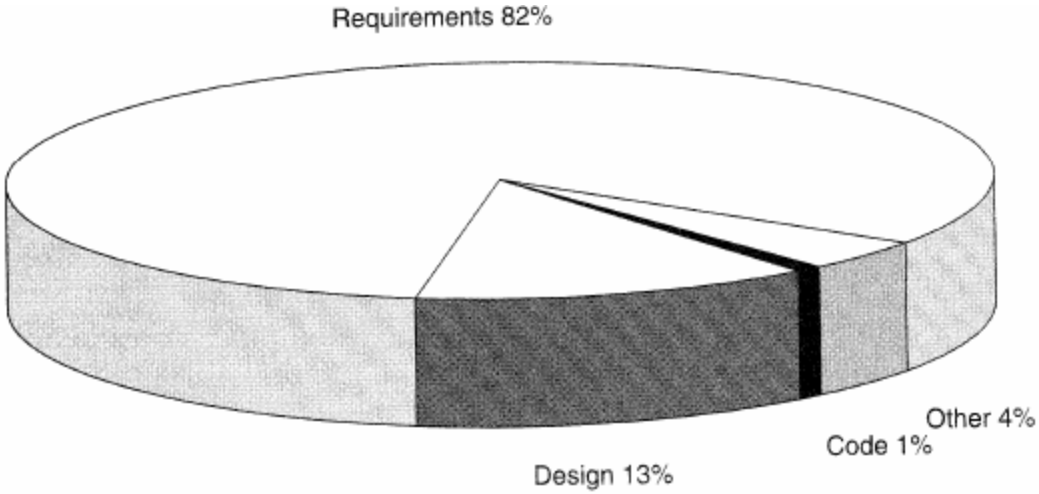
Problems:

- Bags damaged / lost
- System unreliable

https://www.youtube.com/watch?v=cnWVX0Z-OwQ

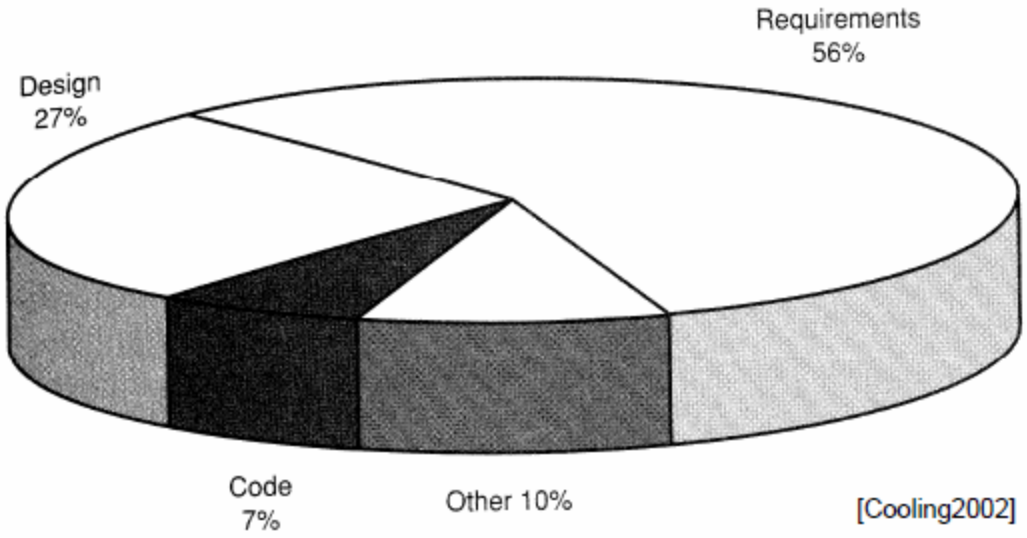
- What do you think went wrong?
- Why did this happen?

5 Why are Requirements so important?



Distribution of software errors

Cost of rectifying errors



[Cooling2002]

Build a system that allows employees to submit vacation requests, and their direct manager — not HR — must approve or reject each request within 24 hours. If no action is taken in 24 hours, the request is automatically approved. The employee receives an email notification for both outcomes.

HOCHSCHULE HAMM-LIPPSTADT

Please describe what you just read to your classmate.

Original Text: Build a system that allows employees to submit vacation requests, and their direct manager — not HR — must approve or reject each request within 24 hours. If no action is taken in 24 hours, the request is automatically approved. The employee receives an email notification for both outcomes.

- who approves (manager, not HR),
- the time limit (24 hours),
- what happens if nobody acts (auto-approve),
- and what triggers a notification (both outcomes)

What is Requirement Engineering?

- **Requirement Engineering (RE)** is the science and discipline concerned with analyzing and documenting requirements.

Requirement:

- (1) A condition or capability needed by a user to solve a problem or achieve an objective.
- (2) A condition or capability that must be met or possessed by a system or system component to satisfy a contract, standard, specification, or other formally imposed documents.
- (3) A documented representation of a condition or capability as in (1) or (2).

[IEEE-Std-610.12-1990]

The inputs and outputs

Input or output	Type	Description
Existing system information	Input	Information about the functionality of systems to be replaced or other systems which interact with the system being specified
Stakeholder needs	Input	Descriptions of what stakeholders need from the system to support their work
Organisational standards	Input	Standards used in an organisation regarding system development practice, quality management, etc
Regulations	Input	External regulations such as health and safety regulations with which the system must comply
Domain information	Input	General information about the application domain of the system
Agreed requirements	Output	A description of the system requirements which is understandable by stakeholders and which has been agreed by them

[Kotonya&Sommerville1998]

RE Elements

a) **Requirement Elicitation** (what is needed)
b) **Requirement Analysis** (clean & check)
c) **Requirement Specification** (documented)
d) **Requirement Validation** (confirm correctness)
e) **Requirement Management** (control changes over time)

[Thayer&Dorfman1997]

a) Requirement Elicitation

- **Requirement Elicitation:** the process through which the customer and developer discover, review, articulate, and understand the users' needs and constraints on the software and development activities
- Requirements elicitation is about discovering what requirements a system should be based upon
- This doesn't involve just asking stakeholders what they **want.** It requires a careful analysis of:
  - The organisation
  - The application domain
  - Organisation processes where the system will be used
- To determine the stakeholders **need.**

[Thayer&Dorfman1997]

Who are they?

- Anyone with a stake in creating or using a new system
  - Hands-on users
  - Their managers
  - The system administrator
  - The client (system owner)
  - The requirements engineer
  - Other Engineers
  - Software designers
  - Programmers

Stakeholders' backgrounds vary: they may:

- come from different departments
- be trained in different disciplines
- have different (possibly conflicting) goals
- be unwilling to consider other stakeholders' goals
- have more or less political influence over requirements decisions

Elicitation Process

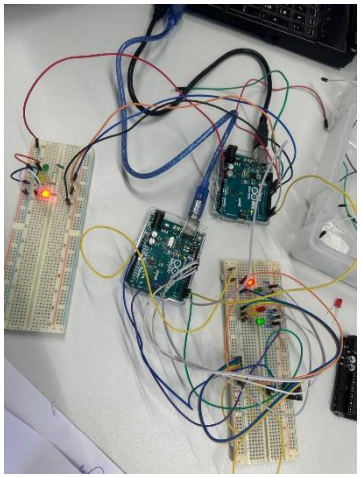
- Establish objectives
  - Business goals
  - Problem to be solved
  - System constraints
- Understand background
  - Organizational structure
  - Application domain
  - Existing systems
- Organize knowledge
  - Stakeholder identification
  - Goal prioritization
  - Domain knowledge filtering
- Collect requirements
  - Stakeholder requirements
  - Domain requirements
  - Organizational requirements

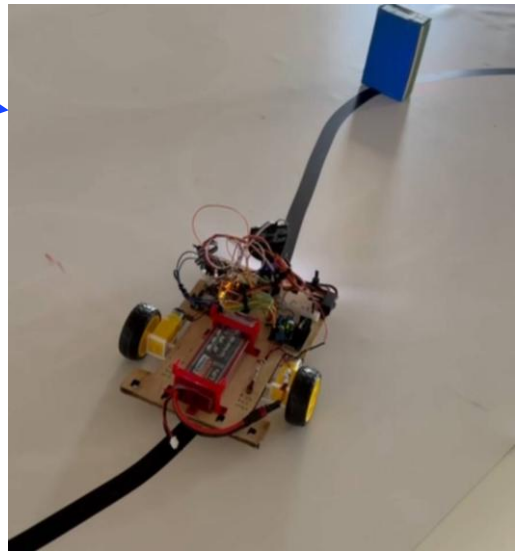
Uncovered Knowledge

- Application domain knowledge
  - E.g. knowledge about airport systems
- Context knowledge
  - E.g knowledge about Denver Airport
- Problem knowledge
  - E.g. knowledge about Denver’s baggage-handling system
- Stakeholders needs and work processes to be supported

Requirements Elicitation Techniques

- Interviews
- Questionnaires
- Examination of documentation
  - Standards
  - Systems manuals
  - Statement of requirements
- Prototyping
- Conversation and interaction analysis





b) Requirement Analysis

- **Requirement Analysis**: the process of analyzing the customers' and users' needs to arrive at a definition of the requirements
- **requirement analysis:**
  - (1) The process of studying user needs to arrive at a definition of system, hardware, or software requirements.
  - (2) The process of studying and refining system, hardware, or software requirements**.**
- Analyze the results of elicitation
  - are the answers consistent?
  - identify trouble spots
  - identify boundaries
  - identify most important requirements
- possibly iterate over elicitation again [IEEE-Std-610.12-1990]

- The analysis has to establish an agreed set of requirements which are **complete**, **consistent**, and **unambiguous**. Such a set can be used as the basis for systems development.
- **Negotiation:** Stakeholders often disagree over requirements. Therefore they need to negotiate to try to reach agreement.

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c) Requirement Specification

- **Requirement Specification**: the development of a document that clearly and precisely records each of the requirements of the software.
- Software Requirement Specification (SRS)
- Requirement : Specific, measurable, and testable

Thayer&Dorfman1997]

"The pedestrian button should work”

"After the button is pressed, the walk signal shall activate within 30 seconds”

Requirement Specification

- Different specification methods have different levels of formality (precision)
- Informal (Plain natural language) , Semi-formal (Structured templates, use cases, tables), Formal (Mathematical notations)
  - the more formal, the more one can precisely state requirements and then verify the implemented system meets the requirements
  - the more formal, the more one might be able to analyze the requirements for consistency, etc.
  - the more formal, typically the more time is required for specification
    - not all projects want to spend a lot of time and effort in writing requirements precisely
    - particularly if requirements will change often

d) Requirement Validation

- **Requirement Validation:** the process of ensuring that the requirement specification is in compliance with the user needs
- system requirements, conforms to document standards, and is an adequate basis for the architectural design.

Did we write down the right requirements?

- **Verification**: chekcs that the code matches the spec
- **Validation**: cheks that the spec matches real human needs

[Thayer&Dorfman1997]

e) Requirement Management

- **Requirement Management**: the planning and controlling of the requirements elicitation, specification, analysis, and verification activities.

[Thayer&Dorfman1997]

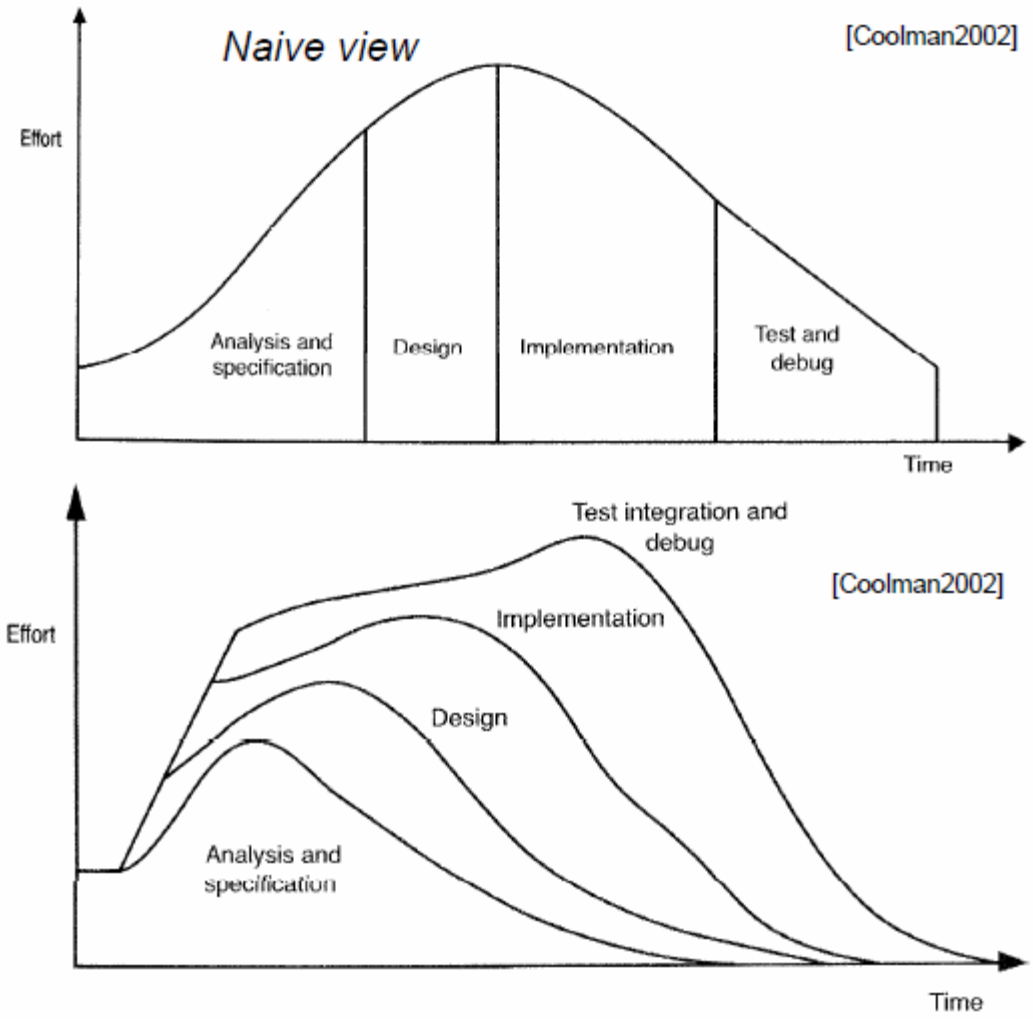
Requirements for Complex Systems

- A system of any but the smallest size will be decomposed into a hierarchy of elements (partitioning):
- This is reflected at the requirement level by:
  - **(1) Allocation**: assigning requirements to elements
  - **(2) Flowdown**: requirements which respond to the allocated high level requirements
  - **(3) Traceability**: keep track of the dependencies

[Thayer&Dorfman1997]

Effort Distribution for Complex Systems

- For complex systems due to allocation and flowdown the Requirement engineering **continues** during the design (and implementation) phase



Requirements vs. Design

Distinction:

[Thayer&Dorfman1997]

- Design solution: **HOW** to achieve something
- Requirements: **WHAT** to achieve

- Requirements and design solutions are sometimes mixed
  - The customer mandates a design solution as a requirement
    - Design solution (HOW): “provide a database for X”
    - Better: ask Why!
  - A derived requirement which is actually a design solution and not a requirement (depends on the perspective)
    - One person’s design is the next person’s requirements

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System modeling language (SysML):

- defines a modeling language for systems engineering applications
- supports the specification, analysis, design, verification and validation of a range of complex systems
  - Includes the possibility of representing system requirements, non-software components (mechanics, hydraulics, sensors, etc.), physical equations, continuous flows (matter, energy, etc.) and allocations
- intended to assist in integrating systems and software methodologies

Who?

- International Council on Systems Engineering (INCOSE) joint with Object Management Group (OMG) in January 2001

Why Model based?

- Improved communications
- Reduced ambiguity
- Reduced errors
- More complete representation
- Enhanced knowledge capture

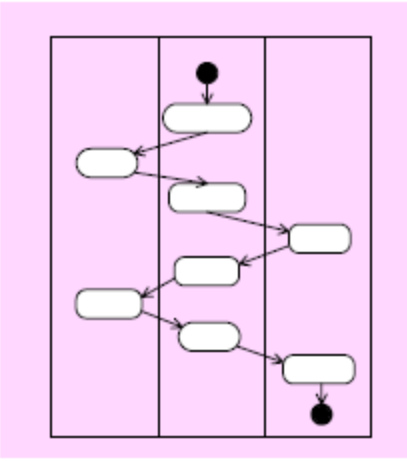
From Document centric to Model centric

Past



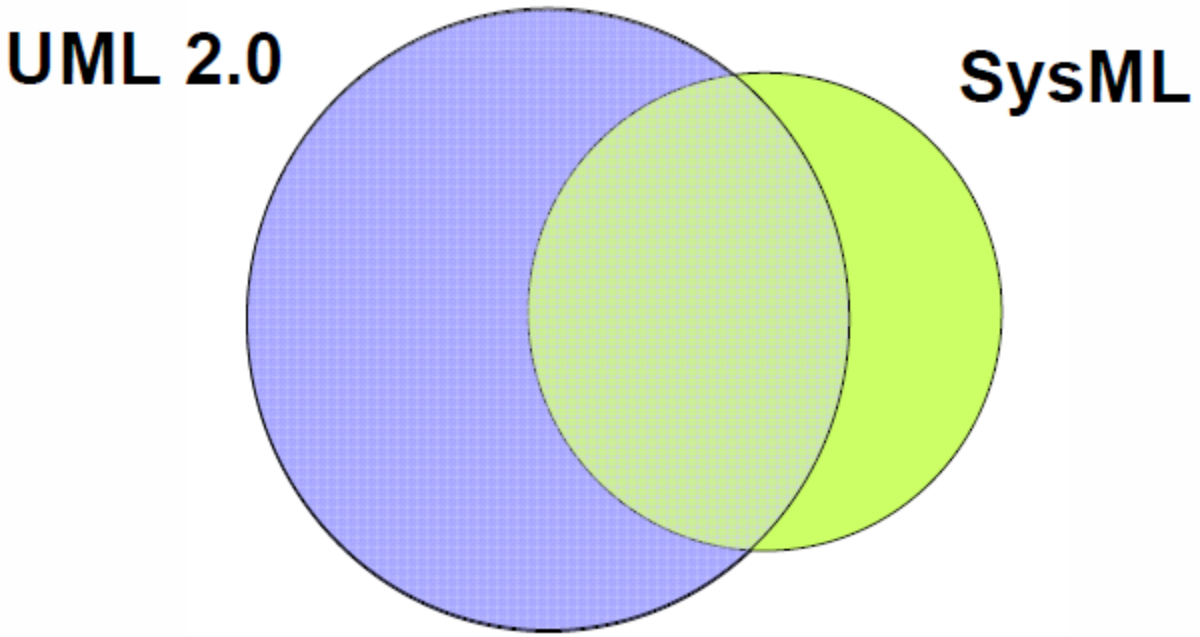
- Specifications
- Interface requirements
- System design
- Analysis & Trade-off
- Test plans

Future



Here: only Requirements!

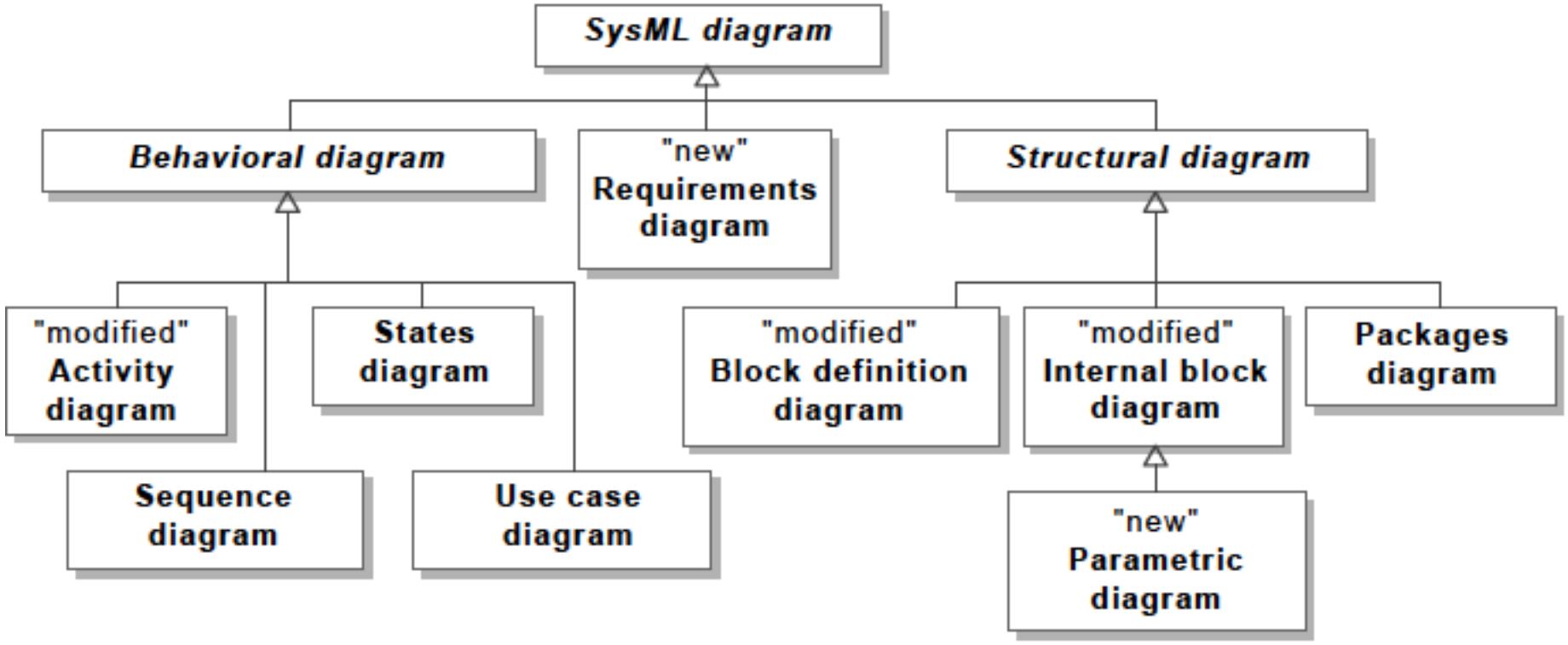
SysML & UML 2.0



Common diagrams: Activities, Block Definitions (UML2::Classes), Internal Blocks (UML2::Composite Structures), Sequences, State Machines, Use Cases

New diagrams: Allocations, Parametric Constraints, Requirements

31 SysML Diagrams



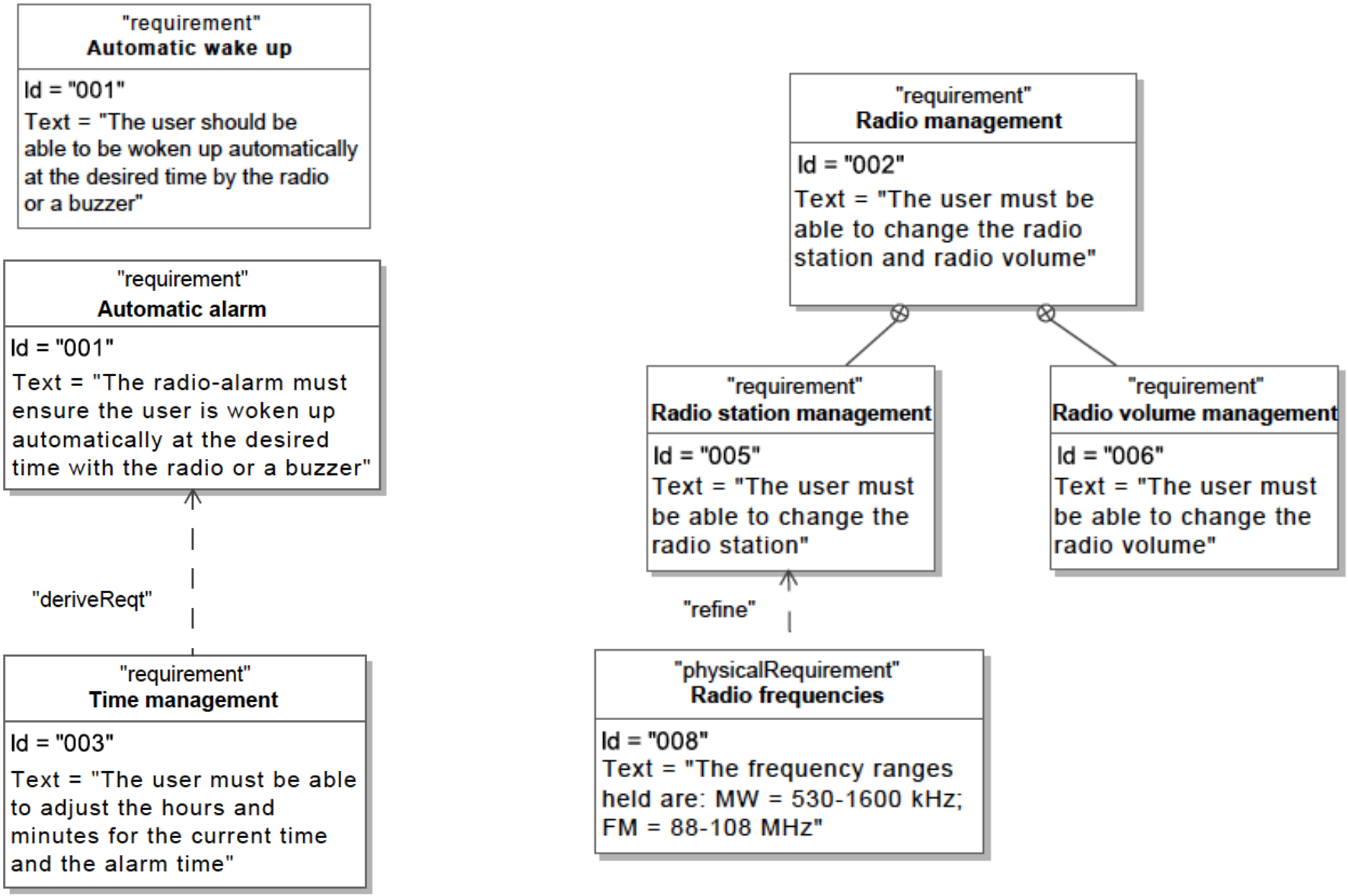
- (1)Requirement Diagrams
- (2)Use Case Diagrams
- (3)Scenarios: Sequence/Activity Diagrams

(1) Requirement Diagram

- offers a graphical representation of the requirements
- Basic properties of a requirement
  - unique identifier
  - textual description of requirement
- Further properties of a requirement
  - priority (e.g. high, medium or low)
  - source (e.g. customer, marketing, technique and legislation)
  - risk (e.g. high, medium or low)
  - status (e.g. suggested, validated, implemented, tested and delivered)
  - verification method (e.g. analysis, demonstration and test)

Requirement Diagram

Example



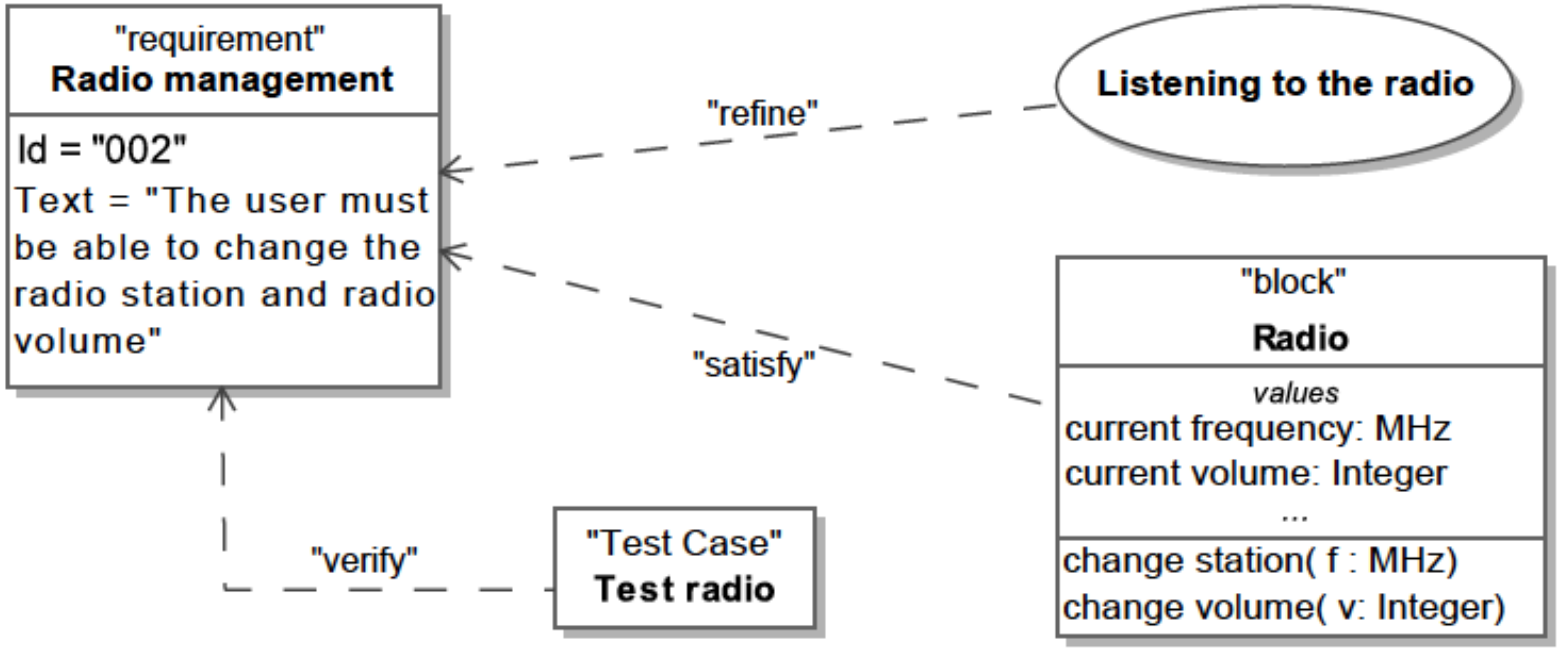
Relations – Requirement diagram specific

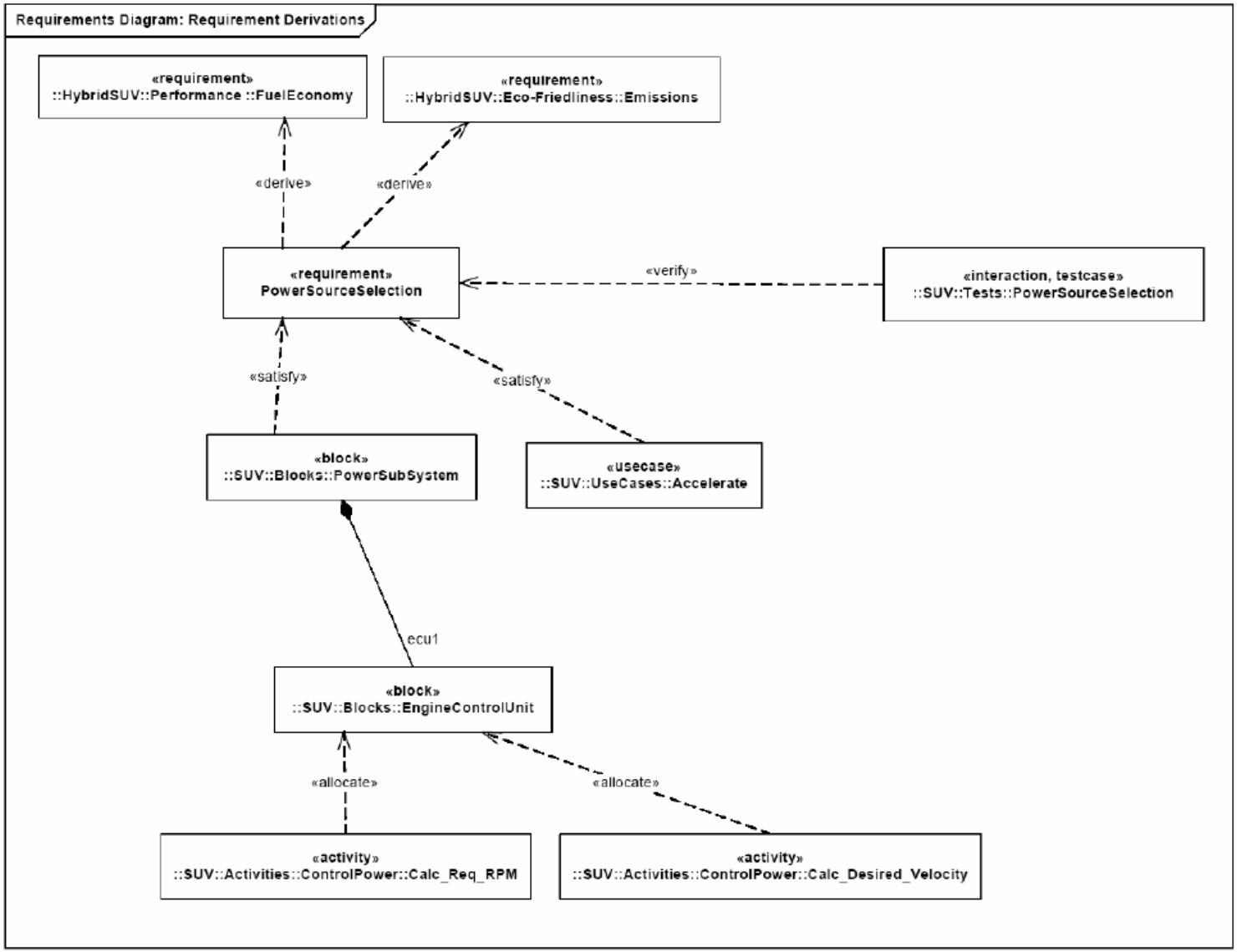
- Containment(): makes it possible to deconstruct a composite requirement into several single requirements, which are then easier to trace with regard to architecture and tests.
- Refinement (refine): consists of adding precision: for example, quantitative data.
- Derivation (deriveReqt): consists of connecting requirements from different levels: for example, connecting system requirements to requirements at the subsystem level. This generally involves making some choices in terms of architecture.

Requirement Diagram

Relations to other SysML Diagrams

- The relation between a requirement and a behavioral element (use case, state diagram, etc.) is denoted by the keyword ***refine***.
- The relation between a requirement and an architecture block is denoted by the keyword ***satisfy***.
- The relation between a requirement and a test case is denoted by the keyword ***verify***.




Requirements Diagram: Requirement Derivations
«requirement»
::HybridSUV::Performance ::FuelEconomy
«requirement»
::HybridSUV::Eco-Friedliness::Emissions
«derive»
«derive»
«requirement»
PowerSourceSelection
«verify»
«interaction, testcase»
::SUV::Tests::PowerSourceSelection
«satisfy»
«satisfy»
«block»
::SUV::Blocks::PowerSubSystem
«usecase»
::SUV::UseCases::Accelerate
ecu1
«block»
::SUV::Blocks::EngineControlUnit
«allocate»
«allocate»
«activity»
::SUV::Activities::ControlPower::Calc_Req_RPM
«activity»
::SUV::Activities::ControlPower::Calc_Desired_Velocity

IV. Discussion and Summary

- Requirements is a **crucial element** of developing systems as about 80% of the software errors are located in the requirements.
- The **system modeling language (SysML)** provides a modeling language for systems engineering applications that supports the specification and analysis of requirements and integrates systems and software methodologies.

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III.7 Bibliography

[Anton+1998] A. Anton et al. The Use of Goals to Surface Requirements for Evolving Systems, International Conference on Software Engineering, pp157-166, IEEE Comp. Soc. Press, 1998

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[IEEE-Std-830-1998] Standards Coordinating Committee of the IEEE Computer Society, The Institute of Electrical and Electronics Engineers, Inc., 345 East 47th Street, New York, NY 10017-2394, USA, IEEE Recommended Practice for Software Requirements Specifications, IEEE Std 830-1998.

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[Sommerville&Sawyer1997] Sommerville, I., and Sawyer, P., Requirements Engineering, John Wiley,1997

[Thayer&Dorfman1997] Software requirements engineering Thayer, Richard H. And Dorfman, Merlin [Hrsg.]:. - Los Alamitos, Calif. [u.a.] : IEEE Computer Soc. Press , 1997 .

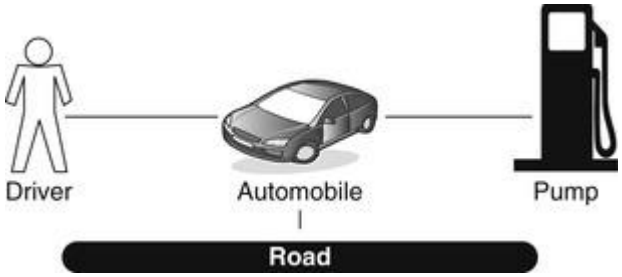
ISBN: 0-8186-7738-4

Automotive Case Study

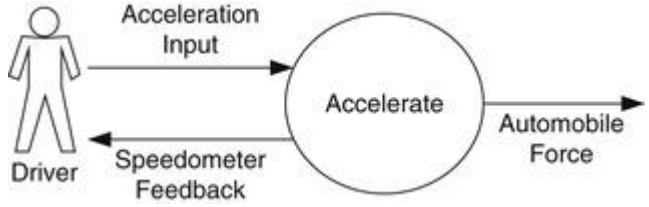
- [1] S. Friedenthal, A. Moore, und R. Steiner, *A practical guide to SysML: the systems modeling language*. Waltham, MA: Morgan Kaufmann, 2012.
  - Chapter 4

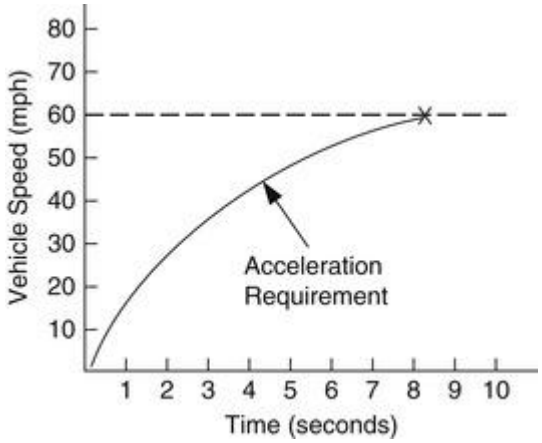
Overview

- Boundary

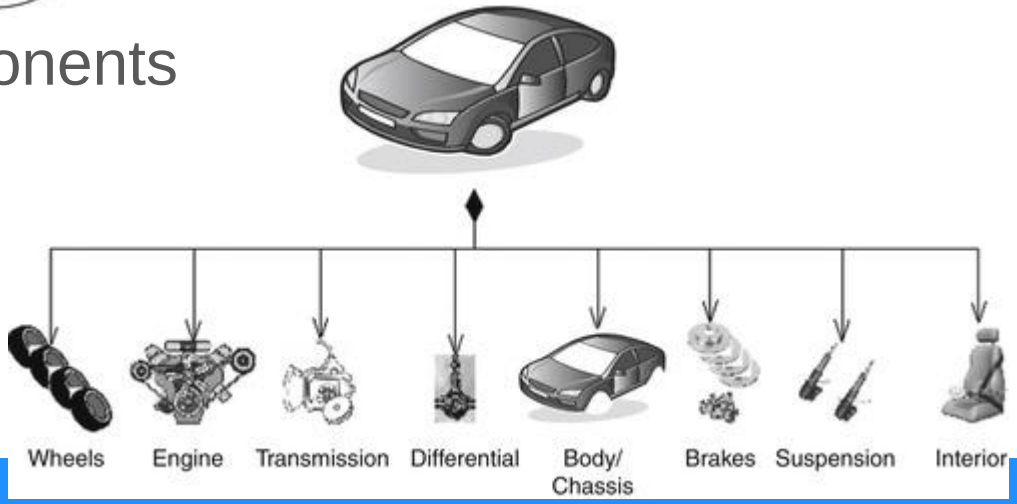


- Functional and performance requirements



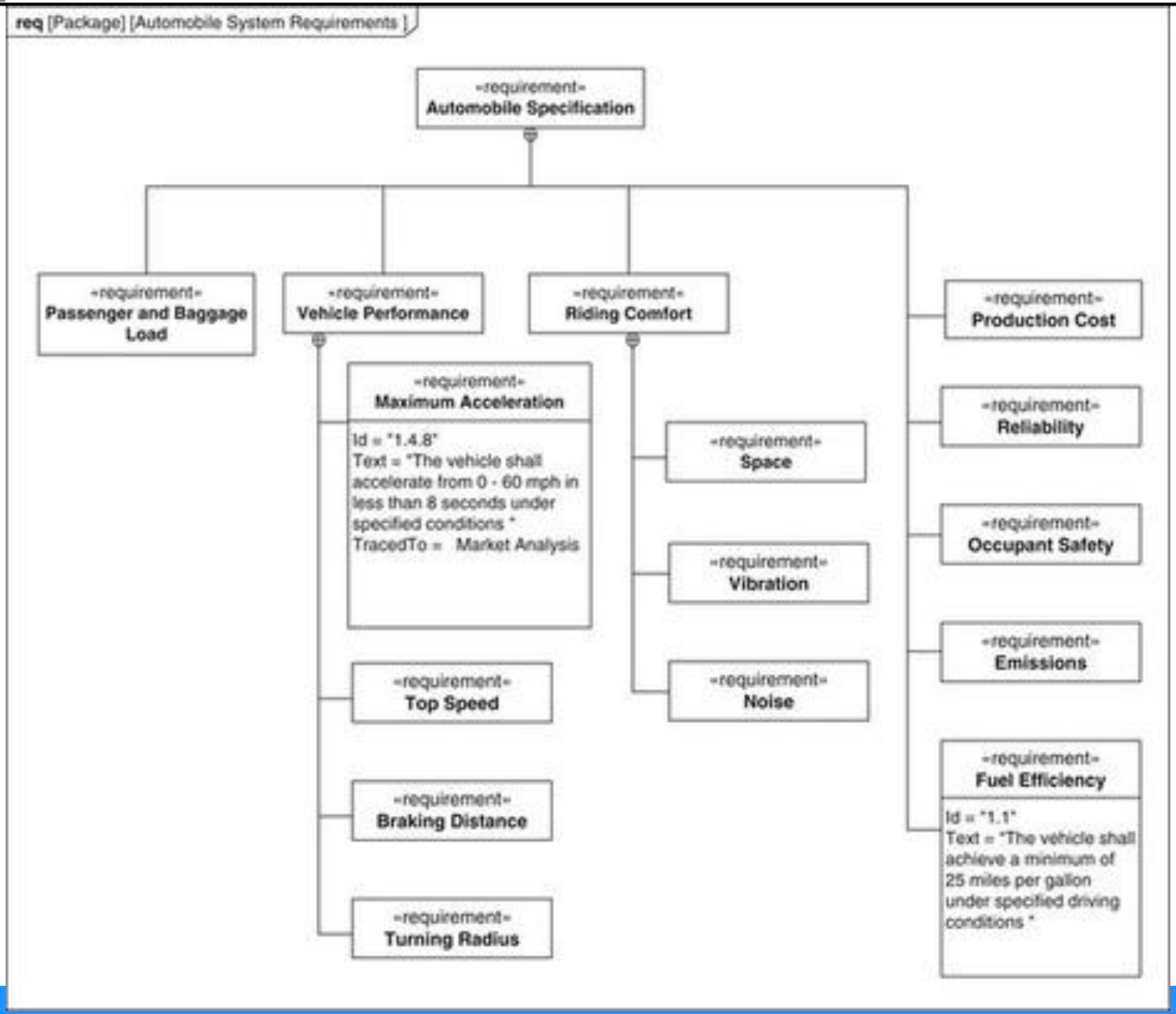


- System components



Automotive Case Study

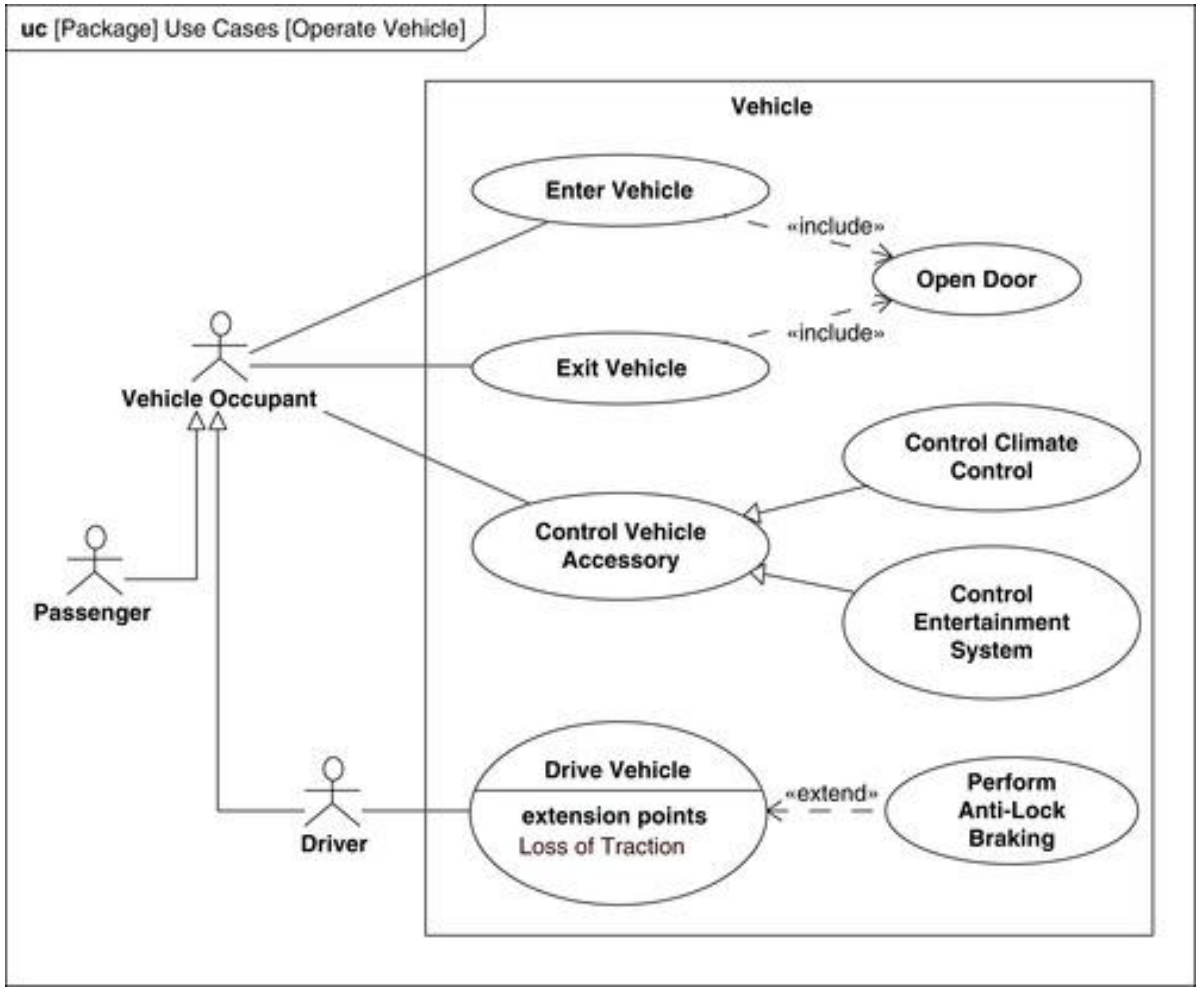
Requirements



Automotive Case Study

Use Cases

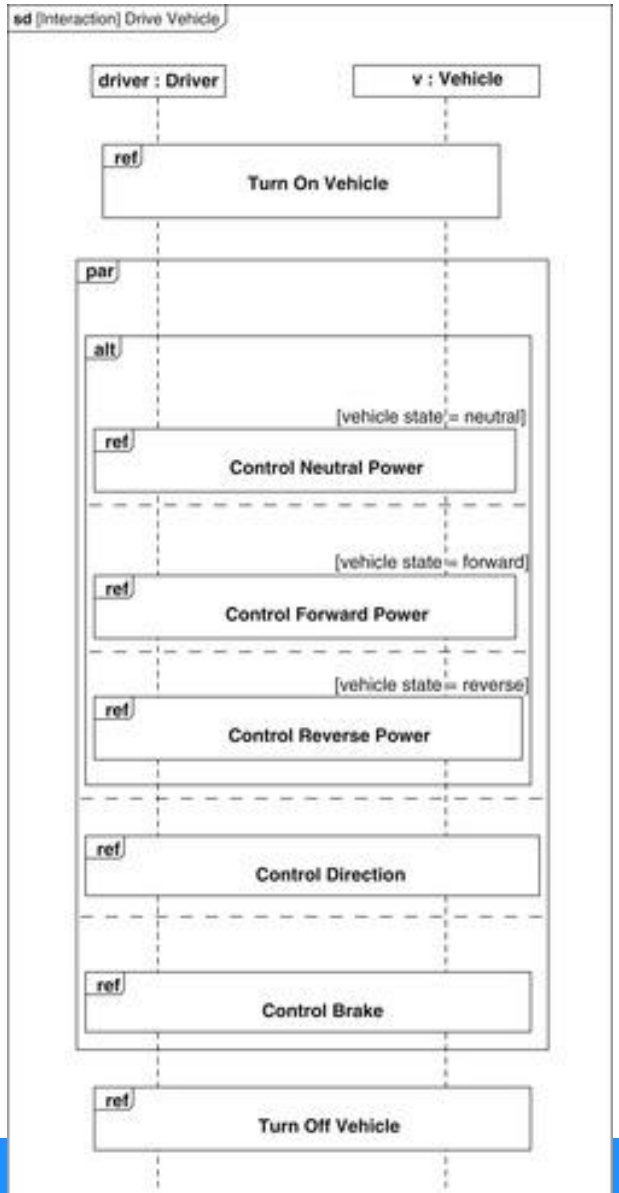
- Functional Requirements



Automotive Case Study

Sequence Diagram

- Interaction between driver and vehicle



Practical work

- Consider your Microcontroller project from last semester
  - Describe the requirements with the help of SysML Requirements
    - Use the reference possibilities of the SysML Requirement Diagram
  - Use a tool for the specification, like draw.io (https://www.diagrams.net/)
- Readings
  - Tim Weilkiens, “Systems Engineering with SysML/UML” (see: https://learning.oreilly.com/library/view/systems-engineering-with/9780123742742/)
    - Chapter 3. UML—Unified Modeling Language
    - 4.1 TO 4.3. THE REQUIREMENT DIAGRAM
    - 4.5. BLOCK DIAGRAMS